> **Leonhard Euler (1707–1783)**
>
> *Euler is the mathematician behind the constant e just defined, among a staggering number of other contributions – yet contemporaries described him as warm and good-natured rather than possessive of his own discoveries. He kept this good nature even after going completely blind in his later years, continuing to produce mathematics of great originality for another two decades – leaving behind so much unpublished work that it kept his assistants busy for four more decades after his death.*

5.3 Sequences and closed intervals

Before moving on to limits of functions, we pause to record two facts about sequences that live inside a closed interval $[a, b]$. Both are purely about sequences (they need nothing beyond the definition of convergence and the notion of a subsequence), and both will be essential tools later, specifically in Chapter 9.

The Bolzano-Weierstrass Theorem

Let $\{x_n\}_{n\in\mathbb{N}}$ be a bounded sequence (i.e., there exist real numbers $a < b$ such that $a < x_n < b$ for all $n \in \mathbb{N}$). Then $\{x_n\}_{n\in\mathbb{N}}$ has a convergent subsequence: there exists a strictly increasing sequence of indices $\{n_k\}_{k\in\mathbb{N}} \subseteq \mathbb{N}$ such that the subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ converges.

Proof. We use a **bisection method** to construct a nested sequence of closed intervals, each containing infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$.

Construction. Set $[a_0, b_0] = [a, b]$; by hypothesis it contains infinitely many terms of the sequence. Given $[a_k, b_k]$ containing infinitely many terms, split it into its left and right halves. At least one of the two halves must still contain infinitely many terms (otherwise $[a_k, b_k]$ itself would contain only finitely many); call that half $[a_{k+1}, b_{k+1}]$ (if both halves qualify, pick either one, say the left one). By construction,

$$[a_0, b_0] \supseteq [a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots, \qquad b_k - a_k = \frac{b-a}{2^k}.$$

The endpoints converge. The sequence $\{a_k\}_{k\in\mathbb{N}}$ is monotone increasing (each interval is nested in the previous one) and bounded above by b, so by the Monotone Convergence Theorem (Theorem 5.1) it converges; call its limit $\bar{x}$. Likewise $\{b_k\}_{k\in\mathbb{N}}$ is monotone decreasing and bounded below by a, so it too converges. Since $2^k \geq k + 1$ for all $k \in \mathbb{N}$ (an easy induction), $2^k \to +\infty$, so for any $\varepsilon > 0$ we can choose k large enough that $\frac{b-a}{2^k} < \varepsilon$; hence $b_k - a_k \to 0$, and $\{b_k\}_{k\in\mathbb{N}}$ converges to the same limit $\bar{x}$.

Extracting the subsequence. Since each $[a_k, b_k]$ contains infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$, we can always choose an index beyond the ones used so far: pick n_0 with $x_{n_0} \in [a_0, b_0]$, and, having chosen $n_0 < n_1 < \cdots < n_{k-1}$, pick $n_k > n_{k-1}$ with $x_{n_k} \in [a_k, b_k]$ (possible precisely because $[a_k, b_k]$ contains infinitely many terms, so not all of them have index $\leq n_{k-1}$). This gives a strictly increasing sequence of indices $\{n_k\}_{k\in\mathbb{N}} \subseteq \mathbb{N}$ and a subsequence

$$x_{n_0}, x_{n_1}, x_{n_2}, x_{n_3}, \ldots, x_{n_k}, \ldots$$

The subsequence converges to $\bar{x}$. Since $x_{n_k} \in [a_k, b_k]$ and $\bar{x} \in [a_k, b_k]$ (being the common limit of a_k and b_k, with $a_k \leq \bar{x} \leq b_k$ for all k), we have

$$|x_{n_k} - \bar{x}| \leq b_k - a_k = \frac{b-a}{2^k} \xrightarrow[k\to\infty]{} 0.$$

Directly from the definition of the limit: given $\varepsilon > 0$, choose K with $\frac{b-a}{2^K} < \varepsilon$; then for all $k \geq K$, $|x_{n_k} - \bar{x}| \leq b_k - a_k \leq \frac{b-a}{2^K} < \varepsilon$. Hence $\lim_{k\to\infty} x_{n_k} = \bar{x}$, which completes the proof. (You will see this same bisection idea again, used to locate a root rather than a limit point, in the proof of Bolzano's Theorem in Chapter 9.) □

> **Bernard Bolzano (1781–1848)**
>
> *Bolzano was a Catholic priest and philosopher who held the chair of philosophy of religion in Prague, and who taught that war and militarism were a waste of society's resources, pressing instead for reform directed toward peace rather than armed conflict between nations. For refusing to recant these and other liberal views, he was formally removed from his professorship on Christmas Eve 1819. Barred from teaching, he spent much of the rest of his life – and much of his own income – supporting poorhouses, homes for the blind, and schools for the poor.*

Example 5.4: The sequence $a_n = (-1)^n$, $n \in \mathbb{N}$, has values

$$1, -1, 1, -1, 1, -1, \ldots, (-1)^n, (-1)^{n+1}, \ldots$$

This sequence is indeterminate. However, if we choose only the even indices we find that a_{n_k} (where $n_k = 2k$) is the subsequence $a_0, a_2, a_4, \cdots = 1, 1, 1, \ldots$ does have the trivial limit of 1. Similarly, the odd indices ($n_k = 2k + 1$) will give us a subsequence with the limit -1.

Remark: The above example demonstrates that a bounded sequence can have more than one convergent subsequence. The Bolzano-Weierstrass theorem tells us that there's *at least* one.

Proposition 5.3 (Convergent sequences in a closed interval)**:** Let $a < b$ and let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence with $x_n \in [a, b]$ for all $n \in \mathbb{N}$. If $\{x_n\}_{n\in\mathbb{N}}$ converges, then

$$\lim_{n\to\infty} x_n \in [a, b].$$

Proof. Let $L = \lim_{n\to\infty} x_n$, and suppose for contradiction that $L > b$. Take $\varepsilon = \frac{L-b}{2} > 0$. By the definition of the limit, there exists $N \in \mathbb{N}$ such that for all $n > N$, $|x_n - L| < \varepsilon$, and in particular

$$x_n > L - \varepsilon = \frac{L+b}{2} > b,$$

contradicting $x_n \leq b$. Hence $L \leq b$. The argument that $L \geq a$ is entirely analogous (take $\varepsilon = \frac{a-L}{2}$ if $L < a$). Together, $L \in [a, b]$. □

We will use both of these results together in Chapter 9, to extract from a sequence confined to a closed interval $[a, b]$ a subsequence whose limit is guaranteed to still lie in $[a, b]$.